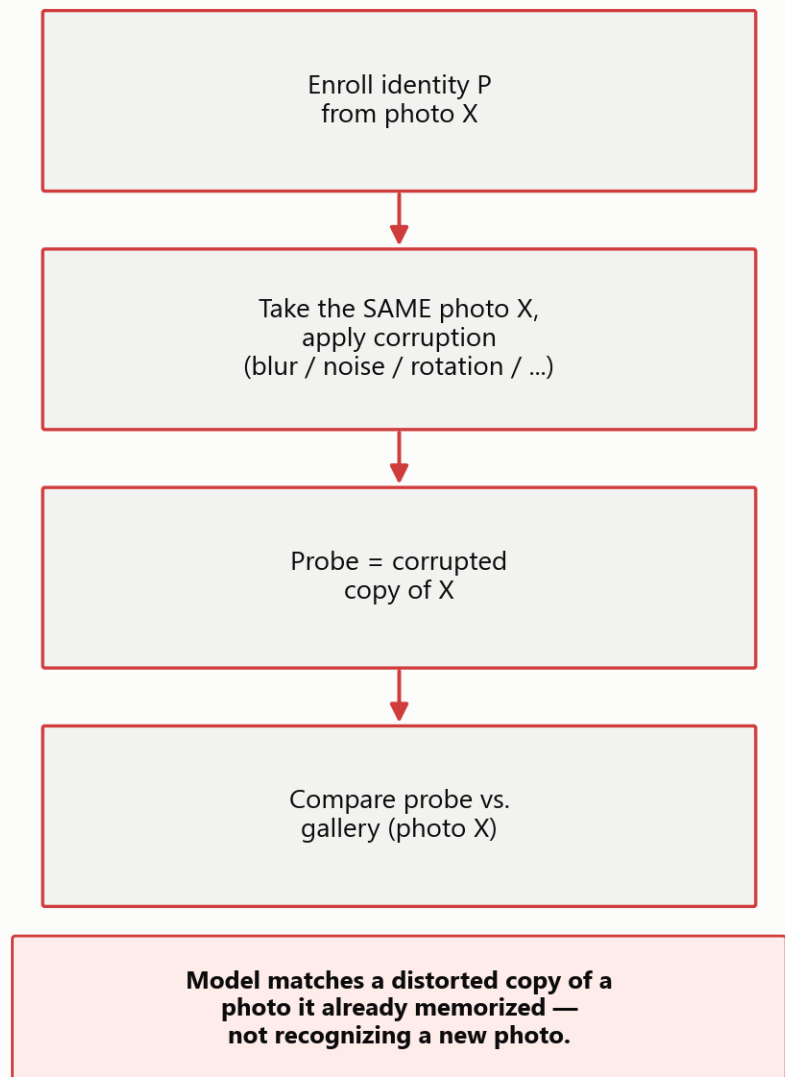
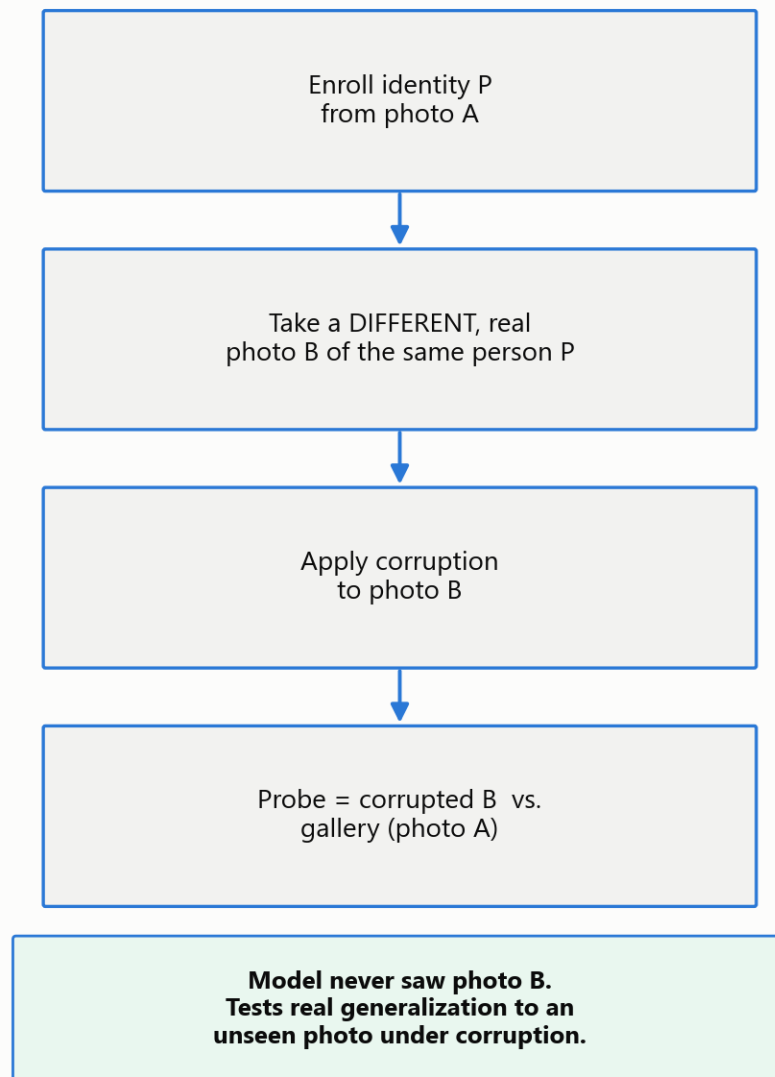


What changed in the LBPH robustness test setup

OLD — same-image protocol (leaky)



NEW — gallery/probe-disjoint protocol



LBPH numbers: old (leaky) vs new (corrected) protocol

*Not the same scale — different protocol, different question.
Old measured memorization; new measures real cross-photo generalization.*

OLD — same-image (transform_sensitivity)

100.00%

clean acceptance

86.66%

41-modification average AR — flagged INVALID

NEW — 1:1 pairwise verification, disjoint photos

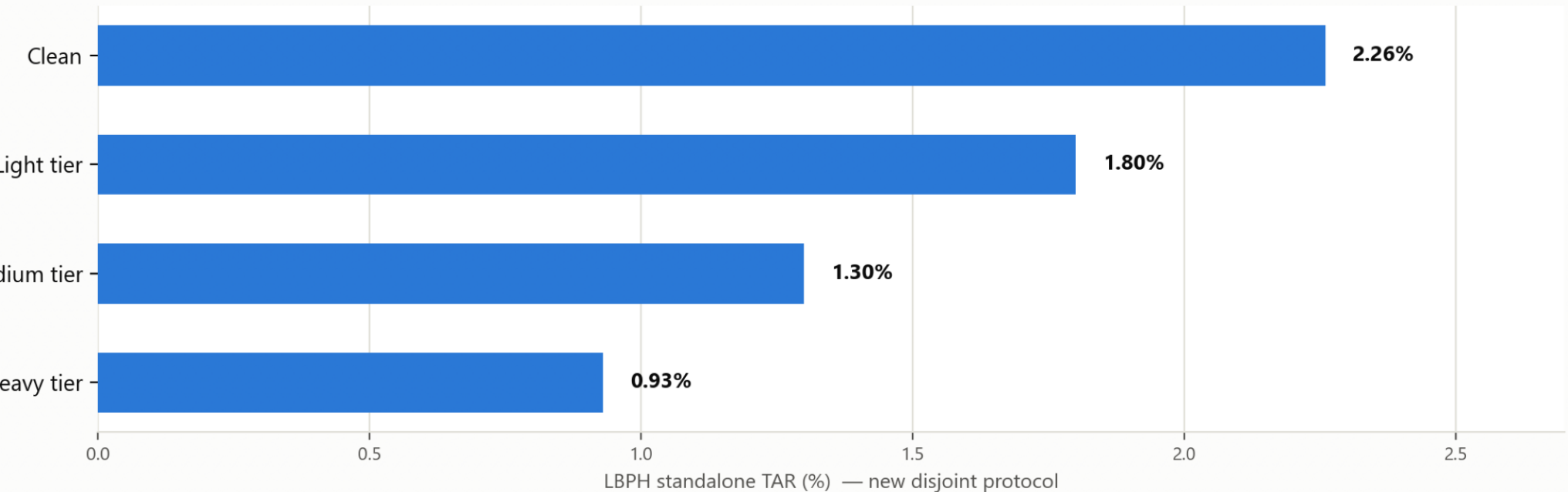
2.26%

clean TAR

~1.41–1.43%

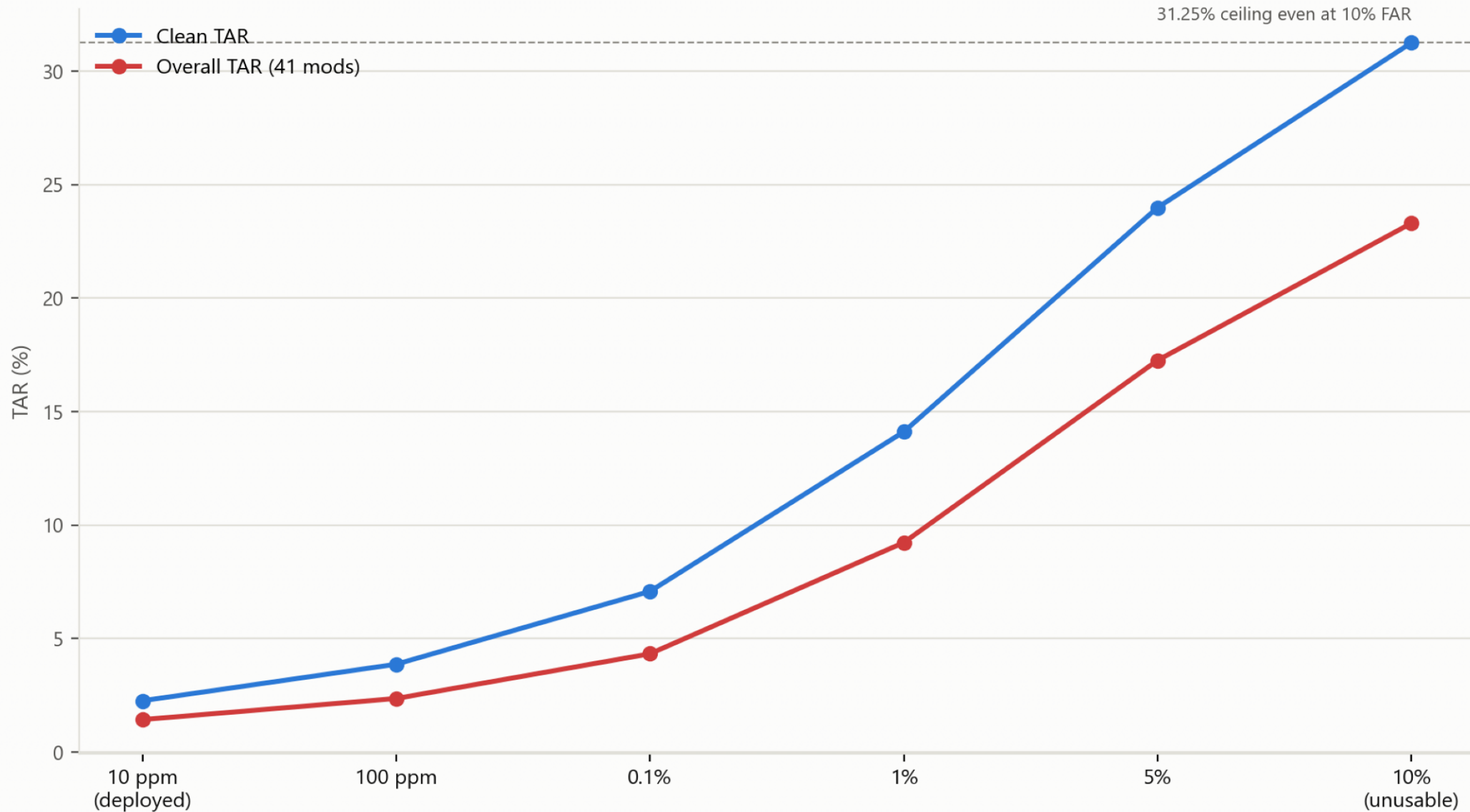
overall 41-mod TAR @ $\tau_{\text{accept}} = 67.03325520645528$

TAR degrades monotonically as modification severity increases



Even giving up FAR almost entirely doesn't rescue LBPH TAR
(genuine/impostor distance overlap, not a threshold-tuning problem)

31.25% ceiling even at 10% FAR



How to read the 41-modification table

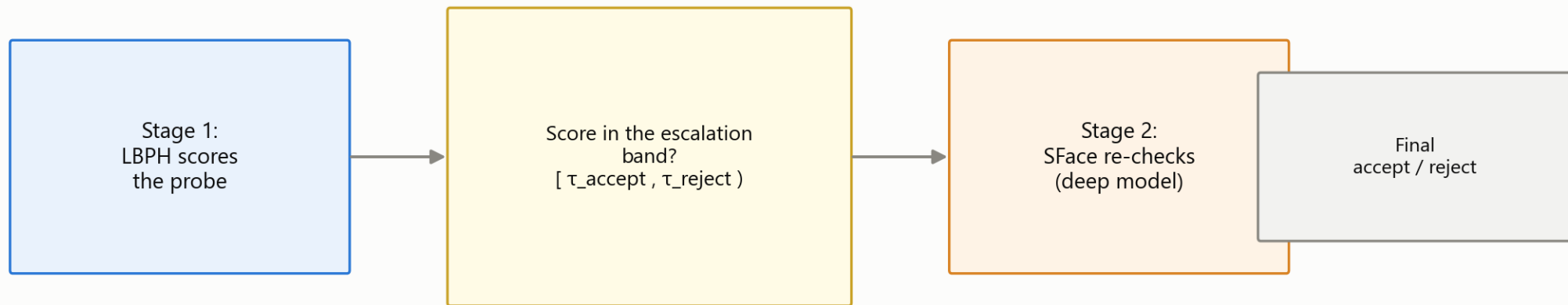
17 modification families x up to 3 severity tiers (Light -> Medium -> Heavy) = 41 rows. 4 columns.

Modification Family	Severity Tier & Detail	LBPH (%)	SFace (%)
Blur	Light (3x3 mask)	51.19%	96.34%
" (Blur)	Medium (5x5 mask)	50.86%	96.16%
" (Blur)	Heavy (7x7 mask)	50.30%	95.65%
Rotation (large-angle)	Heavy (270°)	50.00%	50.00%

4 example rows shown (full 41-row table: pairwise-verification/lfw-results/AR/per_modification_verification_table.png, TAR variant: .../TAR/per_modification_tar_table.png). Rows grouped by family, Light -> Medium -> Heavy within each group.

Note: the LBPH column above is AR (Accuracy@tau), pinned near 50% because FAR = 0% on every row. The TAR breakdown (previous diagram) is the number that actually moves under degradation.

Separate concept from the TAR numbers above — this is the hybrid cascade (1-to-N identification), not the 1:1 verification protocol.



**Escalation % = fraction of probes landing in the band, handed to SFace.
Measured (real 1-to-N identification run, 41-mod suite): 97.51% mean escalation.**